

SAMUDRA

Oil Spill Attribution Engine | SIH26143 (NTRO) | Status as of 11 September 2026

What it is. Not an oil spill detector — detectors already exist. This is an **attribution engine**: given a radar image of a slick and the surrounding ship traffic, it works out **which vessel** is responsible and attaches a probability to the claim.

Build progress	13 of 13 planned stages complete
Automated tests	100 passing, 0 failing
Working incidents	3 (two synthetic, one on real vessel traffic)
Demo runtime	18-41 seconds per incident, fully reproducible from scratch
System layers	7 of 11 fully built; 4 partial or pending

1. Verified results

The system is tested by planting a culprit ship in a simulated scenario with a known release time and position, then checking whether the engine finds it without ever being told the answer. A test scans the source code to prove no part of the pipeline can read the answer key.

Measure	demo-001	demo-002	houston-001
Traffic	Synthetic	Synthetic	REAL - 560 ships
Culprit correctly ranked #1	YES	YES	9th of 12 (see 4.1)
Confidence in top suspect	0.715	0.698	shortlist of 12
Release time error	0.02 h (~1 min)	0.12 h (~7 min)	0.25 h (~15 min)
Release position error	0.98 km	2.34 km	2.53 km
Slick age error	0.02 h	0.12 h	0.50 h
Ships narrowed	24 to 3	46 to 3	426 to 12
Runtime	41 s	41 s	18 s

demo-002 is the important one. It uses a different random seed, a different culprit, twice the traffic and a higher wind band, and the engine was never developed against it. Getting the right answer on a scenario the code has not seen is the only real evidence the physics generalises rather than the constants having been fitted.

Component-level results

Component	Result
Drift model round-trip	IoU 0.86 against ground truth (gate was 0.50)
Ship detection (CFAR)	13 of 13 planted vessels found; 4 false positives
Spoofing / loitering detection	7 of 7 planted anomalies caught, 0 false positives across 47 ships
Oil detection chain	Found the true slick at 0.736 overlap and correctly scored a decoy lower
Evidence integrity	Altering a file, editing the log, or deleting a record are all detected

2. What is built

#	Layer	Status	Notes
1	Data Ingestion	Partial	Real AIS working (NOAA, 560 ships). Satellite/weather fetching not built.
2	Preprocessing & Baseline	Partial	Code complete and correct. Needs 15+ images to start learning; has 1.
3	AI Oil Detection	Partial	Full pipeline built and tested. Model needs one real training run.
4	Vessel Analysis	Partial	Radar ship-detection works. Radar-to-AIS matching not built.
5	AIS Trust & Behaviour	Built	Catches spoofing, loitering, gaps, identity mismatch.
6	Drift Model	Built	Backwards and forwards particle drift through wind and current.
6b	Candidate Pruning	Built	Filters irrelevant traffic and reports the funnel.
7	Attribution Engine	Built	The differentiator. Simulates and ranks. Verified against known answers.
8	Impact & Forecast	Built	+24/48/72h spread with uncertainty cone and shoreline ETA.
9	Investigator Dashboard	Built	Map, funnel, ranked suspects, drift replay, forecast slider.
10	Evidence & Integrity	Built	5-page PDF dossier, MARPOL citations, tamper-proof hash chain.
11	Dissemination & Alerting	Not built	Coast Guard alerting and agency export.

Layer 7 is the differentiator, and it is complete and proven. Everything the project claims to be novel lives there.

3. What is pending

3.1 The oil detector needs one real training run

This is the closest item to done and the most visible. The whole detection chain is built and tested: land masking, wind gating, image tiling, a DeepLabv3+ neural network, polygon extraction, and confidence fusion. What is missing is a trained model. The only checkpoint that exists came from a two-minute test run on 96 images, which existed purely to prove the training loop works. The system knows this and **refuses to use those weights**, falling back to a known-good input rather than quietly producing numbers nobody can defend.

- **Effort:** roughly 2 hours on the available NVIDIA RTX 4050 GPU.
- **Blocker:** PyTorch was installed as the CPU-only build and needs reinstalling with GPU support. A separate network restriction blocks the download of standard starting weights.
- **What it unlocks:** the accuracy table (confusion matrix) proving the detector separates oil from background.
- **Note:** neither training dataset labels look-alikes, so the result will be oil-versus-background only. That is a limit of the available data, not a shortcut.

3.2 The system does not yet remember past incidents

Two of the project's five claimed advantages are about improving over time, and both are built but unseeded. The baseline layer learns what each patch of sea normally looks like under given wind conditions, so a genuine spill stands out and a known natural seep does not. The code accumulates correctly but needs at least fifteen satellite images before the statistics mean anything, and there is one — so it runs in fallback mode and says so on every run. Separately, the repeat-offender history is wired into the scoring formula but always reads zero, because the database that would record confirmed offences was not built. A repeat offender is currently treated exactly like a first-timer.

3.3 The strongest anti-spoofing check is not implemented

Ships broadcast their own position over an unencrypted radio signal with no authentication whatsoever, so that signal can simply be faked. The most powerful way to catch a liar is to compare it against radar: if a ship claims to be somewhere and the radar shows empty water there, that is hard evidence. Radar ship-detection is built and finds 13 of

13 planted vessels, but the step that pairs radar contacts with ship identities was not built, so the DARK and PHANTOM classifications are never actually computed from imagery.

3.4 Data has to be supplied manually

Ship tracking data is genuinely real — a full day of NOAA broadcast points covering 560 vessels, with real reporting gaps and missing fields. Everything else is handed to the system rather than fetched by it: there is no automatic Sentinel-1 satellite download, no Sentinel-2/3 optical branch for filling revisit gaps, and no live weather service call, so wind and current fields are simulated.

3.5 No alerting output

The system produces the evidence dossier but does not yet push alerts to the Indian Coast Guard under NOS-DCP, and has no severity-tiered notification or agency data export.

4. Known limitations (findings, not unfinished work)

4.1 In a single-lane shipping channel, no method can name one ship

On the real Houston data the culprit ranks ninth rather than first. This is not a bug, and it was measured rather than assumed. Across all 97 candidate vessels in that waterway, the **maximum** separation between any ship and the nearest other moving ship is **72 metres**. Several tugs run the identical line within the hour, so the oil slick each would have produced is geometrically indistinguishable — every candidate's direction matches within 1 to 13 degrees, against 45 to 86 degrees for the synthetic decoys. Geometry genuinely cannot separate them.

What the engine does there is still useful: it narrows 426 vessels to a shortlist of 12 containing the culprit, and pins the release to within 15 minutes. The system reports ranked candidates with probabilities and never issues a single accusation — which is exactly the behaviour the design calls for. **Presenting this honestly is stronger than showing a forced first place.**

4.2 Shoreline impact is untested against a real landfall

The coastline map is a simplified built-in of the Indian west coast, because the network in use blocks the download of the proper dataset. Neither demo scenario drifts ashore within 72 hours, so the shoreline arrival-time feature has never been exercised against an actual impact. The system correctly reports 'not assessed' for areas the bundled coastline does not cover, rather than printing a meaningless distance.

4.3 The slick-age cross-check is not independent

Age is estimated from the winning hypothesis and cross-checked against a physical spreading law. That second estimate requires an assumed discharge volume and oil type, so it is a consistency check rather than independent corroboration. The dossier states this. On the Houston incident the two estimates correctly report **disagreement**, because the assumed constant was calibrated for open-ocean discharges and does not fit a harbour slick.

5. Recommended next steps, in priority order

#	Task	Effort	Value
1	Reinstall PyTorch with GPU support and run the full model training	~2.5 h	Unlocks the accuracy table, the one missing headline number
2	Obtain 15+ satellite scenes over one area to seed the baseline	Data-bound	Activates the 'learns over time' advantage
3	Build radar-to-AIS matching (Hungarian assignment)	~3 h	Enables DARK/PHANTOM detection, the strongest anti-spoofing signal
4	Build the repeat-offender database (connector A)	~3 h	Closes the feedback loop; system compounds across incidents

#	Task	Effort	Value
5	Add Coast Guard alerting and export (layer 11)	~3 h	Completes the Disaster Management framing
6	Add automatic satellite and weather ingestion	~1 day	Removes manual data handling

6. Running the demo

All three incidents are pre-built on disk; nothing is computed live. Start the dashboard and switch incidents from the dropdown. Hard-refresh the browser first.

Start the dashboard	<code>.venv/Scripts/python -m samudra.api --port 8000</code>
Then open	<code>http://127.0.0.1:8000</code>
Rebuild one incident	<code>bash scripts/run_demo.sh demo-001 --regen --no-serve</code>
Run the test suite	<code>.venv/Scripts/python -m pytest -q</code>

7. Data provenance

Data	Source	Real or simulated
Ship tracking (AIS)	NOAA MarineCadastre, Houston, 31 Dec 2025	Real
Radar training images	Deep-SAR SOS (6,455 labelled pairs)	Real
Radar training images	CSIRO Sentinel-1 set, DOI 10.25919/4v55-dn16	Real
Radar scene for the demo	Generated with realistic speckle and look-alikes	Simulated
Wind and current	Generated with realistic statistics	Simulated
The oil release itself	Planted, so correctness can be verified	Simulated

The originally planned Zenodo radar dataset was unavailable during the build — a documented multi-day outage of Zenodo's own services, not a data-access failure at our end. Two independent open datasets were substituted. The problem statement explicitly permits simulated AIS; using real traffic instead is a strengthening of the brief, not a deviation.